Gaussian Approximation for Asynchronous Q-learning



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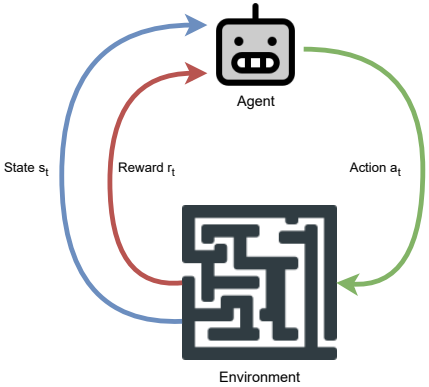
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Reinforcement learning: basic quantities

A Markov decision process is a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$.

- $\mathcal{S}$: state space; $s_t \in \mathcal{S}$.
- $\mathcal{A}$: action space; $a_t \in \mathcal{A}$.
- $P(s' \mid s, a)$: transition law.
- $R(s, a) = r(s, a)$: immediate reward.
- $\gamma \in (0, 1)$: discount factor.
- $\pi(a \mid s)$: policy, giving the distribution of actions at state s.



$$V^{\pi}(s) = \mathbb{E}^{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \,\middle|\, s_0 = s\right].$$

Goal: find a policy $\pi^\star$ maximizing $V^{\pi}(s)$ for every initial state s.

Discounted Markov decision processes

An MDP is specified by $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$. Under a policy π,

$$a_t \sim \pi(\cdot \mid s_t), \qquad s_{t+1} \sim P(\cdot \mid s_t, a_t).$$

Assumption A1 (Finite discounted MDP)

The state and action spaces are finite, $d = |\mathcal{S}||\mathcal{A}|$. Rewards are deterministic, $r(s,a) \in [0,1]$, and $0 < \gamma < 1$.

The action-value function is

$$Q^\pi(s,a) = \mathbb{E}^\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \;\middle|\; s_0 = s,\ a_0 = a \right].$$

Goal: estimate $Q^\star(s,a) = \sup_\pi Q^\pi(s,a)$ when P is unknown.

The Bellman optimality equation

The Bellman operator is

$$(\mathcal{T}Q)(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a)[\max_a Q(s',a)] \,.$$

It is a contraction:

$$\left\| \mathcal{T}Q - \mathcal{T}\widetilde{Q} \right\|_\infty \leq \gamma \left\| Q - \widetilde{Q} \right\|_\infty .$$

Hence $Q^\star = \mathcal{T}Q^\star$ is the unique fixed point, and

$$\pi^\star(s) \in \arg\max_a Q^\star(s,a).$$

Q-learning replaces the Bellman expectation by a sampled transition [11].

Asynchronous Q-learning

We observe one trajectory generated by a fixed behavior policy π_b. At time t, only the visited coordinate is updated:

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha_t \delta_t,$$
$$\delta_t = r(s_t, a_t) + \gamma \max_a Q_t(s_{t+1}, a) - Q_t(s_t, a_t).$$

All other coordinates remain unchanged.

- π_b determines which state-action pairs are observed.
- The maximum targets the optimal policy, which may differ from π_b.
- Observations are dependent and visits need not be uniform.

Two regimes of data generation

	Synchronous	Asynchronous
Data source	Generative model	One trajectory under π_b
One iteration	One sample for each (s, a)	One observed transition
Update	All d coordinates	One coordinate

Let $\mu(s, a)$ be the stationary visitation frequency of (s, a). Exploration requires

$$\mu_{\min} = \min_{s,a} \mu(s, a) > 0.$$

One synchronous iteration uses d transitions; an asynchronous one uses one.

Averaging and statistical inference

With polynomial stepsizes, consider the Polyak–Ruppert average

$$\alpha_t = \frac{c_0}{(t+k_0)^\omega}, \quad \frac{1}{2} < \omega < 1, \qquad \bar{Q}_n = \frac{1}{n}\sum_{t=1}^{n} Q_t.$$

Under suitable assumptions,

$$\sqrt{n}\,(\bar{Q}_n - Q^\star) \Rightarrow N(0, \Sigma_\infty).$$

- Averaging gives a $\sqrt{n}$ scale for a fixed MDP.
- Gaussian quantiles can calibrate confidence intervals for $Q^\star$.
- A quantitative CLT measures the approximation error at finite n.

The averaging principle goes back to Polyak–Juditsky [7].

Why approximation over rectangles?

For axis-aligned rectangles $\mathcal{R}_d = \{\prod_{j=1}^{d}(a_j, b_j]\}$, define

$$d_{\mathcal{R}}(X, Z) = \sup_{R \in \mathcal{R}_d} \Big| \mathbb{P}(X \in R) - \mathbb{P}(Z \in R) \Big|.$$

Let $Z \sim N(0, \Sigma_\infty)$ and let $c_{1-\eta}$ be the $(1-\eta)$-quantile of $\|Z\|_\infty$.

$$d_{\mathcal{R}}\Big(\sqrt{n}(\bar{Q}_n - Q^\star), Z\Big) \leq \delta_n$$

implies the simultaneous coverage guarantee

$$\left| \mathbb{P}\left(\left\| \bar{Q}_n - Q^\star \right\|_\infty \leq \frac{c_{1-\eta}}{\sqrt{n}} \right) - (1-\eta) \right| \leq \delta_n.$$

This directly controls all state-action coordinates at once.

Existing results

Averaged Q-learning

- Li et al. (2023): functional CLT and statistical efficiency in the synchronous setting [4].
- Liu (2026): asynchronous Q-learning, with a $\widetilde{O}(n^{-1/6})$ Wasserstein bound [5].

High-dimensional Gaussian approximation

- Chernozhukov–Chetverikov–Kato: rectangle bounds for independent sums [1].
- Quantitative martingale and TD results use different covariance conditions and metrics; we return to them in the proof.

Our question: a rectangle CLT for asynchronous Q-learning.

Assumptions: exploration and mixing

The behavior policy induces

$$K_b(s, s') = \sum_a \pi_b(a \mid s) P(s' \mid s, a).$$

Write η_b for its invariant law and $\mu(s, a) = \eta_b(s)\pi_b(a \mid s)$.

Assumption A2 (Exploration and mixing)

Every pair is visited: $\mu_{\min} = \min_{s,a} \mu(s, a) > 0$. For some $\tau_{\text{mix}} < \infty$,

$$\sup_s \left\| K_b^t(s, \cdot) - \eta_b \right\|_{\text{TV}} \leq (1/4)^{\lfloor t/\tau_{\text{mix}} \rfloor}.$$

Assumptions: stepsizes and the optimality gap

Assumption A3 (Polynomial stepsizes)

With $a = (1-\gamma)\mu_{\min}$ and $c_0, k_0 > 0$, take

$$\alpha_t = \frac{c_0}{(t+k_0)^\omega}, \quad \frac{1}{2} < \omega < 1, \quad c_0 \le \frac{1}{2a}, \quad \alpha_0 \le 1, \qquad k_0^{1-\omega} \ge \frac{8}{ac_0}.$$

Assumption A4 (Unique optimal action)

There is a unique optimal action $\pi^\star(s)$ in each state, and

$$\kappa = \min_s \min_{a \neq \pi^\star(s)} \{Q^\star(s, \pi^\star(s)) - Q^\star(s,a)\} > 0.$$

Why the optimality gap matters

Assumption A4 separates the optimal action from all other actions:

$$Q^\star(s, \pi^\star(s)) - Q^\star(s, a) \geq \kappa > 0, \qquad a \neq \pi^\star(s).$$

If $\|Q - Q^\star\|_\infty < \kappa/2$, then

$$Q(s, \pi^\star(s)) - Q(s, a) \geq \kappa - 2\|Q - Q^\star\|_\infty > 0.$$

Thus small estimation errors do not change the greedy policy.

Consequently,

$$(\mathcal{T}Q - \mathcal{T}Q^\star)(s, a) = \gamma \sum_{s'} P(s' \mid s, a)\Big[Q(s', \pi^\star(s')) - Q^\star(s', \pi^\star(s'))\Big].$$

This linearization is the starting point for the Gaussian approximation. At a tie, arbitrarily small perturbations can switch the maximizing action, and the local behavior of the maximum can remain nonlinear.

Bellman noise and the limiting covariance

At the optimum, the Bellman noise is

$$\varepsilon_t = e_{s_t,a_t}\{r(s_t,a_t) + \gamma V^\star(s_{t+1}) - Q^\star(s_t,a_t)\}.$$

Its stationary covariance is diagonal:

$$\Sigma_\varepsilon = \mathrm{diag}_{s,a}\Big\{\gamma^2 \mu(s,a)\,\mathrm{Var}_{P(\cdot|s,a)}[V^\star(s')]\Big\}.$$

Assumption A5 (Nondegenerate Bellman noise)

$\mathrm{Var}_{P(\cdot|s,a)}[V^\star(s')] > 0$ for every (s,a), so $\Sigma_\varepsilon \succ 0$.

With $D_\mu = \mathrm{diag}\{\mu(s,a)\}$,

$$G = D_\mu(I - \gamma P^{\pi^\star}), \qquad \Sigma_\infty = G^{-1}\Sigma_\varepsilon G^{-\top}.$$

Here $P^{\pi^\star}$ follows a transition and then the optimal action.

Gaussian approximation for Q-learning

Theorem 1

Under A1–A5, initialize $0 \le Q_0 \le (1-\gamma)^{-1}$ *and let* $Z \sim N(0, \Sigma_\infty)$. *Then*

$$d_{\mathcal{R}}\Big(\sqrt{n}(\bar{Q}_n - Q^\star), Z\Big) = \widetilde{O}_{\mathrm{MDP}}\Big(n^{-(1-\omega)/2} + n^{-(\omega-1/2)} + n^{-1/4}\Big).$$

In particular, $\omega = 2/3$ *gives*

$$\boxed{d_{\mathcal{R}}\Big(\sqrt{n}(\bar{Q}_n - Q^\star), Z\Big) \lesssim \frac{\log^4 nd}{n^{1/6}}}.$$

$\widetilde{O}$ suppresses logarithms in n, d. Constants depend on mixing, visitation probabilities, the gap and covariance [9], but not on d.

Proof idea: an asymptotically linear representation

Let $\Delta_t = Q_t - Q^\star$ and let $\mathcal{F}_{t-1}$ contain the trajectory up to s_t. The Bellman equation gives

$$\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0.$$

Summing the Q-learning recursion yields

$$\sqrt{n}\,(\bar{Q}_n - Q^\star) = \underbrace{\frac{1}{\sqrt{n}} \sum_{t=0}^{n-1} G^{-1}\varepsilon_t}_{W_n} + G^{-1}R_n.$$

1. Bound the nonlinear and averaging remainder R_n.
2. Approximate the martingale sum W_n by a Gaussian vector.
3. Transfer the approximation through the remainder.

Controlling the Q-learning remainder

The moment estimates give, for $p \geq 1$ and $t \geq 1$,

$$\left(\mathbb{E}\,\|\Delta_t\|_\infty^p\right)^{1/p} \lesssim_{\mathrm{MDP}} p\log(2dt^2)\,t^{-\omega/2}.$$

The optimality gap controls the nonlinear Bellman remainder:

$$\left\|\mathcal{T}Q - \mathcal{T}Q^\star - \gamma P^{\pi^\star}(Q-Q^\star)\right\|_\infty \leq \frac{4\gamma}{\kappa}\,\|Q-Q^\star\|_\infty^2\,.$$

Consequently, up to logarithmic factors,

$$\|R_n\|_{L^p(\ell_\infty)} \lesssim_{\mathrm{MDP},p} n^{-\omega/2} + n^{-(1-\omega)/2} + n^{-(\omega-1/2)}.$$

The first term is dominated by $n^{-(1-\omega)/2}$ for $\omega > 1/2$.

Martingale CLT

Write $C(s) = \mathbb{E}[\varepsilon_t \varepsilon_t^\top \mid s_t = s]$. For $X_t = G^{-1}\varepsilon_t/\sqrt{n}$,

$$V_t = \mathbb{E}[X_t X_t^\top \mid \mathcal{F}_{t-1}] = \frac{1}{n} G^{-1} C(s_t) G^{-\top}.$$

- $C(s_t)$ depends on the random current state.
- It is supported on actions at s_t, so $\operatorname{rank}(V_t) \le |\mathcal{A}|$.
- Individual covariances can be singular even when the accumulated covariance is nondegenerate.

We need a rectangle CLT based on accumulated covariance. An independent-sum theorem or a positive lower bound on every V_t does not directly cover this setting.

Martingale assumptions and notation

$$S_n = \sum_{k=1}^{n} X_k, \quad V_k = \mathbb{E}[X_k X_k^\top \mid \mathcal{F}_{k-1}], \quad \Sigma_n = \sum_{k=1}^{n} \mathbb{E} V_k.$$

Assumption M1 (Martingale differences and moments)

$X_k \in \mathbb{R}^d$ is $\mathcal{F}_k$-measurable, $\mathbb{E}[X_k \mid \mathcal{F}_{k-1}] = 0$ and $\mathbb{E}\,\|X_k\|_\infty^3 < \infty$.

Assumption M2 (Deterministic total predictable variation)

$$\sum_{k=1}^{n} V_k = \Sigma_n \quad \text{a.s.}, \qquad \min_j (\Sigma_n)_{jj} > 0.$$

Write $P_k = \sum_{i=k}^{n} V_i$. Individual V_k may be random and singular.

A Gaussian approximation theorem for martingales

Theorem 2 (Deterministic predictable variation)

Assume M1–M2 and let $Z_n \sim N(0, \Sigma_n)$. For every deterministic matrix $H \succ 0$,

$$d_{\mathcal{R}}(S_n, Z_n) \lesssim \frac{\log^{5/4}(ed)}{\underline{\sigma}(\Sigma_n)^{1/2}} \left[\overline{\sigma}(H) + \sum_{k=1}^{n} \mathbb{E} \frac{\|X_k\|_\infty^3}{\lambda_{\min}(P_k + H)} \right]^{1/2}.$$

Here $\underline{\sigma}^2(M) = \min_j M_{jj}$ and $\overline{\sigma}^2(M) = \max_j M_{jj}$.

The denominators involve **remaining covariance**, $P_k + H$. The matrix H regularizes the final part of the sum.

This is Theorem 1 of the paper [9].

Comparison with Kojevnikov–Song (2022)

Kojevnikov–Song work conditionally on $\mathcal{F}_0$ and set

$$B_k = \mathbb{E}[X_k X_k^\top \mid \mathcal{F}_0], \qquad \lambda_*^2 = \min_k \lambda_{\min}(B_k) > 0.$$

Their general lower-orthant bound includes a discrepancy term for $V_k - B_k$ [3].

- If $V_k = B_k$, their bound gives $\widetilde{O}(n^{-1/4})$ under controlled standardized moments.
- Our theorem permits singular B_k and uses $P_k + H$ instead of a positive lower bound for each summand.
- We control the total predictable variation, with a separate extension when it is random.

The gain is in covariance assumptions and applicability.

Singular increments can accumulate full covariance

Let $n = 2m$ and let ξ_j, η_j be independent, centered, unit-variance variables with third absolute moments at most M. In $\mathbb{R}^2$, set

$$X_{2j-1} = \sqrt{\frac{2}{n}}\,\xi_j e_1, \qquad X_{2j} = \sqrt{\frac{2}{n}}\,\eta_j e_2.$$

Then every V_k has rank one, while $\Sigma_n = I_2$. With $H = I_2/n$,

$$\lambda_{\min}(P_k + H) \geq \frac{n-k+1}{n}.$$

The martingale theorem therefore yields

$$d_{\mathcal{R}}(S_n, N(0, I_2)) \lesssim_M n^{-1/4}\sqrt{1+\log n}.$$

Other martingale approximation results

Röllin (2018); Srikant (2025)

Stein–Lindeberg arguments give Wasserstein bounds, first for scalar martingales and then for vector martingales and TD learning [8, 10].

Wu–Wei–Rinaldo (2025)

Their vector martingale theorem also uses cumulative covariance and deterministic total variation. It controls Wasserstein distance. Their Markov setting yields convex-distance bounds [12].

Present theorem

A rectangle-distance bound with ℓ_∞ moments and explicit logarithmic dimension factors. The choice of metric is central.

Random predictable variation in Q-learning

The actual Q-learning martingale does not satisfy M2. Instead,

$$\langle W \rangle_n = \frac{1}{n} \sum_{t=0}^{n-1} G^{-1} C(s_t) G^{-\top}$$

concentrates around $\Sigma_n^\nu = \mathrm{Var}(W_n)$.

- Stop the martingale and add a Gaussian increment to make its total predictable variation deterministic.
- Control the modification and covariance tails using mixing.

Together these give $d_{\mathcal{R}}(W_n, N(0, \Sigma_n^\nu)) = \widetilde{O}_{\mathrm{MDP}}(n^{-1/4})$ [6, 9].

Proof of the martingale theorem

Gaussian smoothing. For $R \in \mathcal{R}_d$, use

$$\varphi_{R,\epsilon}(x) = \mathbb{P}(x + \epsilon Z_0 \in R), \qquad Z_0 \sim N(0, I_d).$$

Lindeberg replacement. Replace one increment at a time. Under M2, the remaining covariance is predictable:

$$P_k = \Sigma_n - \sum_{i<k} V_i \quad \text{is } \mathcal{F}_{k-1}\text{-measurable.}$$

Stein's equation. Bounds on the smoothed Stein solution introduce $(P_k + H)^{-1}$ and third ℓ_∞ moments.

Optimize the smoothing scale using Gaussian anti-concentration. The argument combines Röllin, Kojevnikov–Song and Fang–Koike [2, 3, 8].

Completing the Q-learning proof

Gaussian anti-concentration transfers the bound from W_n to $W_n + G^{-1}R_n$. Up to logarithms, the three terms are

$$\begin{aligned} \text{martingale approximation:} &\quad n^{-1/4}, \\ \text{averaging and boundary terms:} &\quad n^{-(1-\omega)/2}, \\ \text{nonlinear and Markov remainders:} &\quad n^{-(\omega-1/2)}. \end{aligned}$$

Balancing the last two terms gives

$$\frac{1-\omega}{2} = \omega - \frac{1}{2} \quad \Longrightarrow \quad \omega = \frac{2}{3}, \qquad d_{\mathcal{R}} = \widetilde{O}_{\mathrm{MDP}}(n^{-1/6}).$$

Conclusion: quantitative Gaussian approximation for joint inference in Q-learning. Feasible confidence regions additionally require covariance estimation or a justified bootstrap procedure.

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